MBB: HOLLOW DESIGN ACCEPTED

```
cshell design accepted: J = +0.4958 (optimisation grid), +0.4954 (shifted grid) > ln(1 + g_min) = 0.0198,
feasible, export mass-matched and watertight

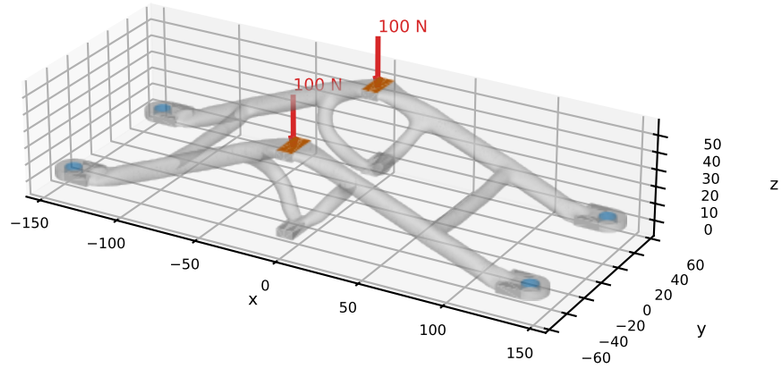
mass      target 88.64 g   final 88.63 g   error -1.5370e-04
optimiser grid  J 0.4958 (weighted)  ln K/K0 0.1073  K/K0 1.113  P/P0 2.421  governing buckling  feasible True   (accept: J >
    ln 1.02 = 0.0198, K/K0 >= 0.85, P/P0 >= 0.85)
shifted grid    J 0.4954 (weighted)  ln K/K0 0.1065  K/K0 1.112  P/P0 2.421  governing buckling  feasible True   (accept: J >
    ln 1.02 = 0.0198, K/K0 >= 0.85, P/P0 >= 0.85)

export    h 0.2 mm  faces 4747668  watertight True  volume 8.522e+04 mm3  material bodies 1  shells 3  enclosed voids 2
    escape holes 0
wall [mm] min 1.492  p05 1.502  median 2.486  max 27.03
closed_area_frac   0.03875
hollow_area_frac   0.7547

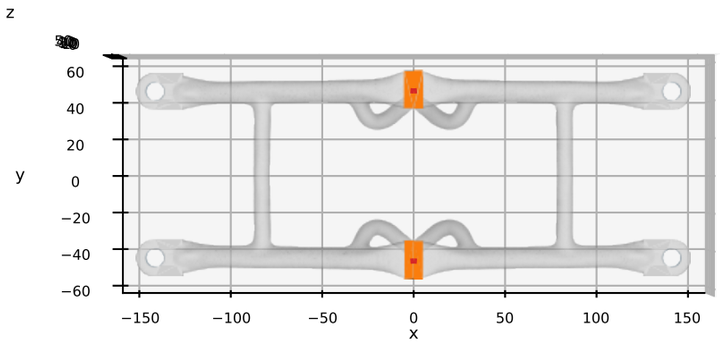
optimiser: {'method': 'mma', 'starts': ['solid'], 'max_iter': [60, 80, 40], 'move': 0.1, 'tol_x': 0.005, 'tol_f': 0.0001,
    'mass_tol': 0.0001, 'sliver_weight': [0.0, [0.0, 0.05], 0.1], 'seed': 0}
grid: {'phases': [{'h_e': 0.8, 'h_g': 0.2}, {'h_e': 0.4, 'h_g': 0.1}], 'export_h_mm': 0.05, 'rho_min': 1e-06,
    'tessellation_mm': 0.02, 'active_margin_mm': 0.5, 'geometry_device': 'auto', 'geometry_offload': True}
growth: {'enabled': True, 'phase': 0, 'tol': 0.001, 'patience': None, 'quantiles': [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
    0.9, 1.0], 'step_tol': 0.0001, 'max_iters_per_step': 25, 'min_iters_per_step': 2, 'samples': 16}
objective: {'type': 'weighted', 'constraints': [], 'constraint_scale': 0.01, 'feasibility_tol': 0.001, 'w_stiffness': 0.5,
    'w_capacity': 0.5, 'min_gain': 0.02, 'floor_stiffness': 0.85, 'floor_capacity': 0.85, 'floors_as_constraints': True}
export cfg: {'h_mm': 0.2, 'device': 'gpu', 'closed_thr': 0.9, 'escape_holes': False}
```

MBB: supports (blue, all DOF fixed; 4 patches) and loads (orange faces; 2 red arrows, one per loaded patch at its area centroid, length and label = its share of the total 200 N). Faces read from bc_faces.npz (as analysed).

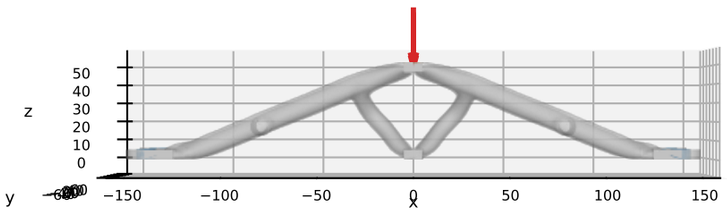
isometric



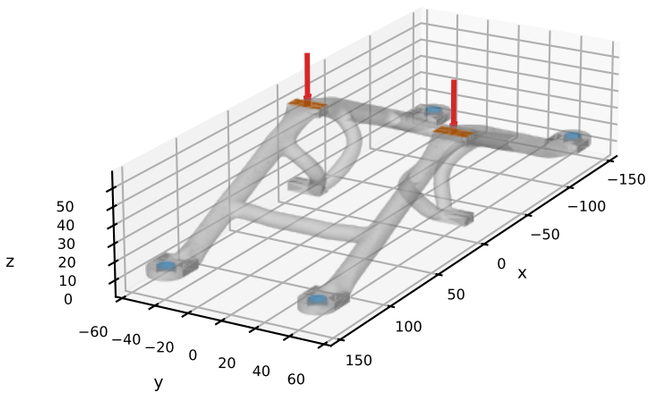
top (+z)



front (-y)

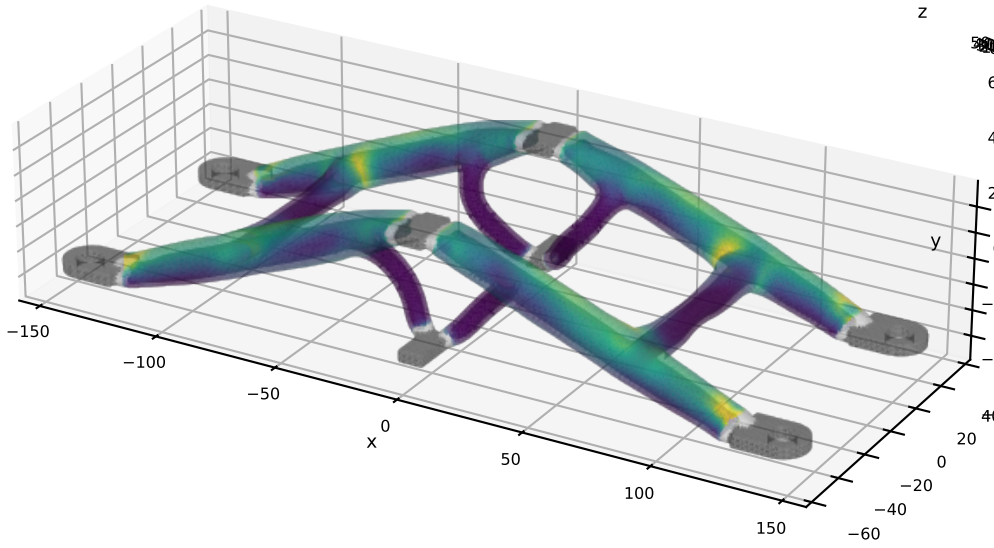


rear isometric

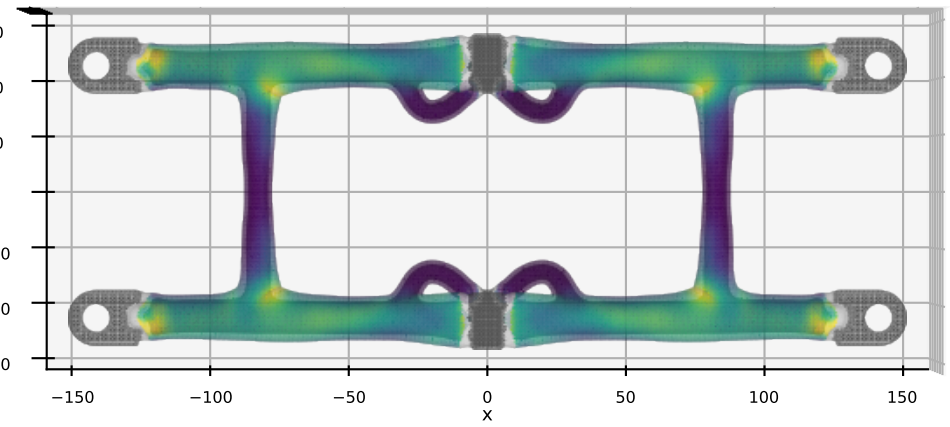


MBB: final design (STL 4,747,668 faces, shown decimated to 173,665; wall thickness from fields.vtp)

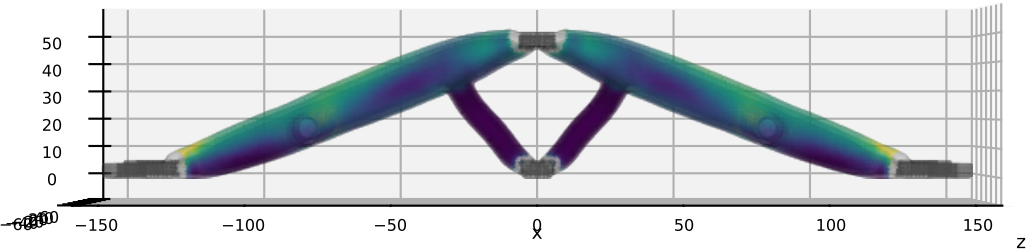
isometric



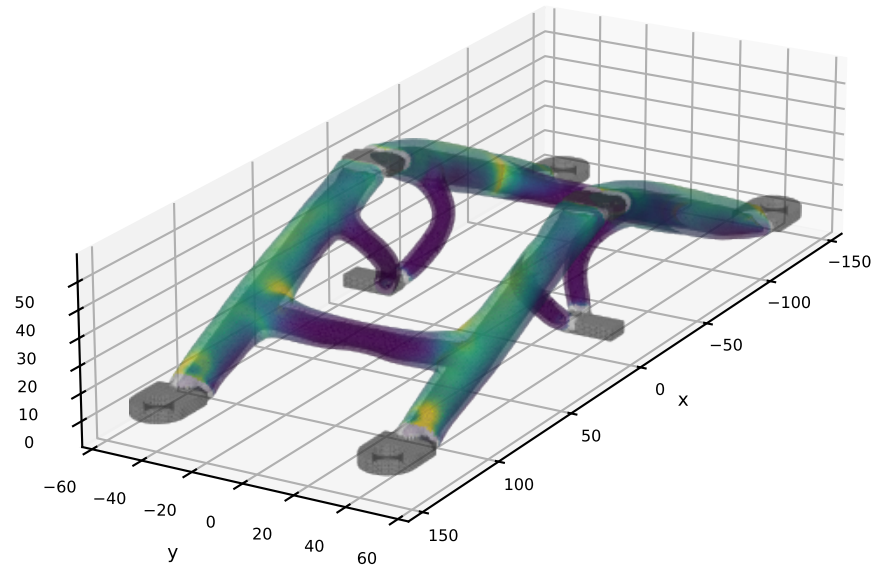
top (+z)

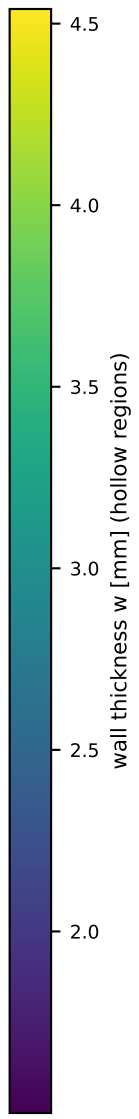


front (-y)



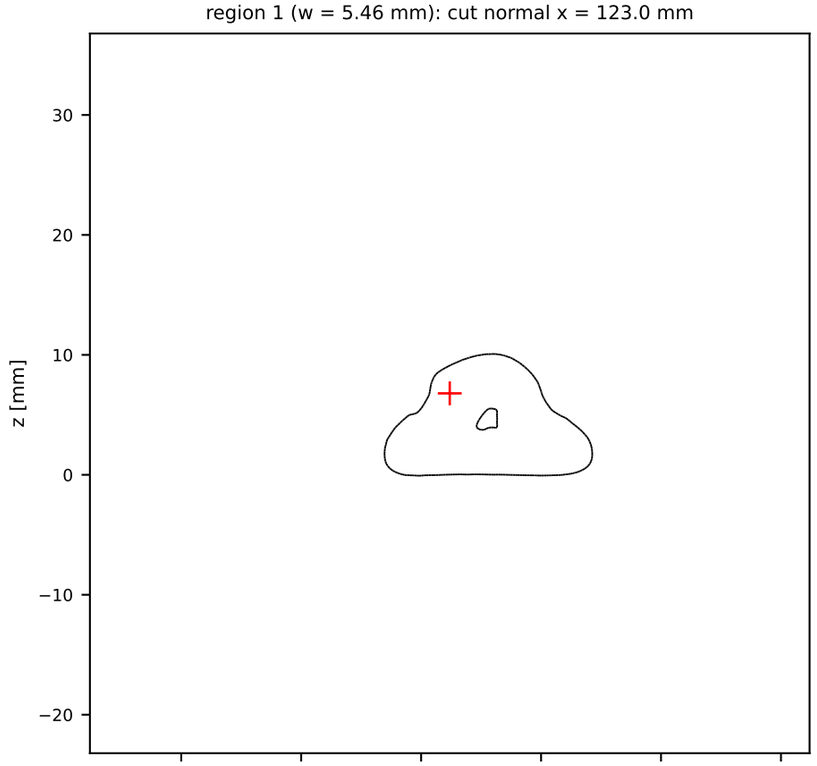
rear isometric

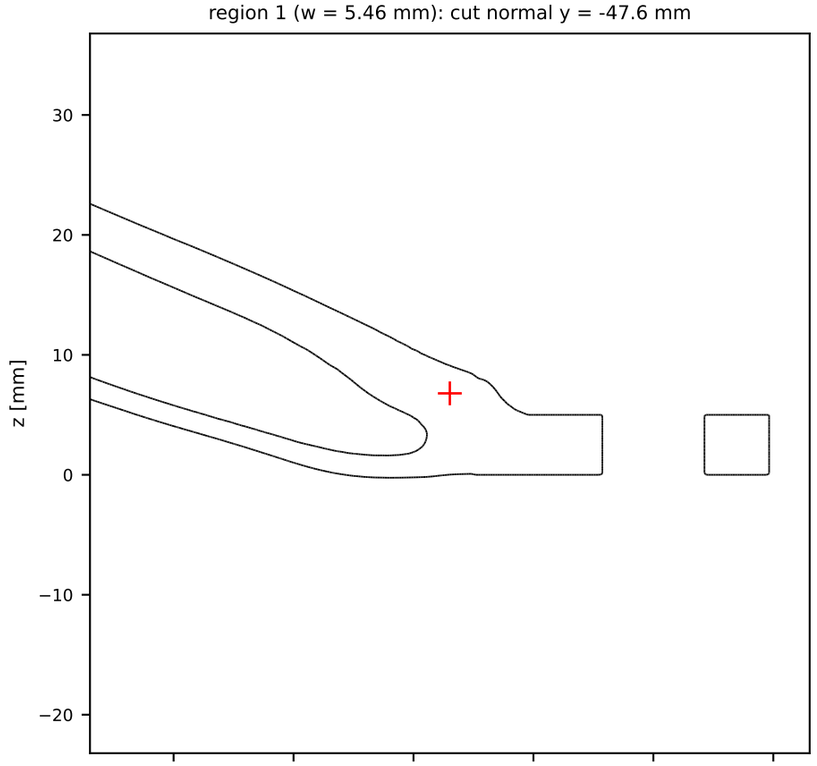


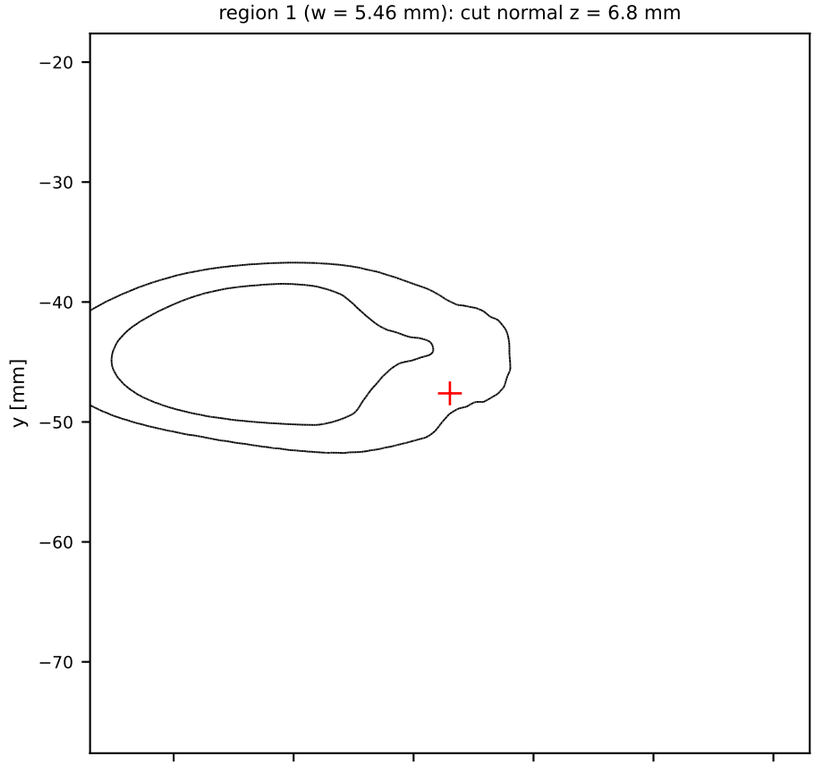


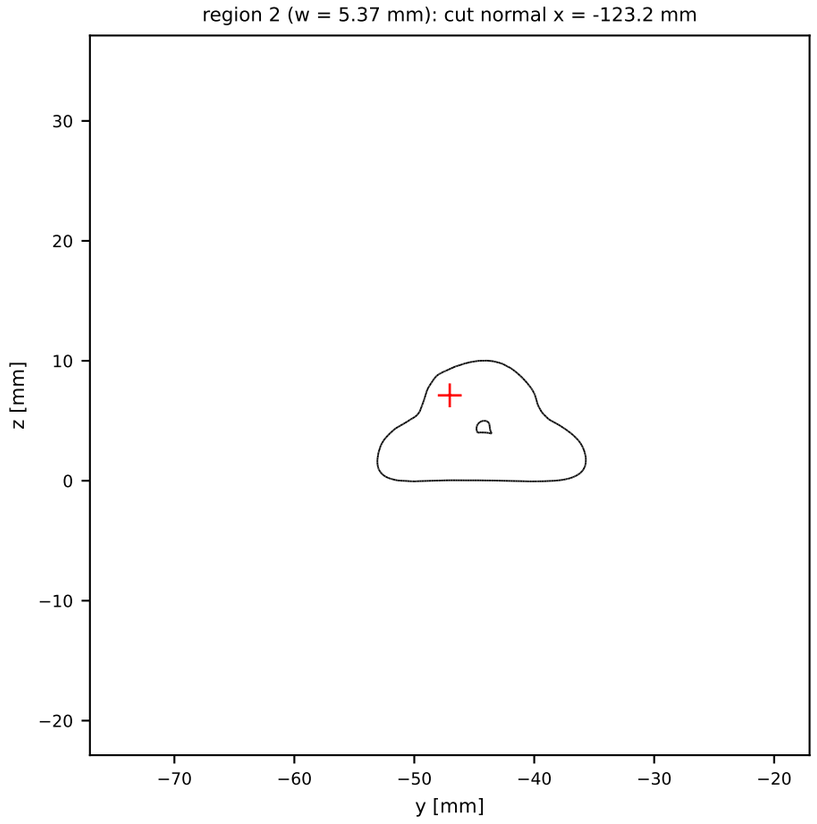
light grey = solid (closed) regions; dark grey = mounts; the colourbar covers the hollow regions only

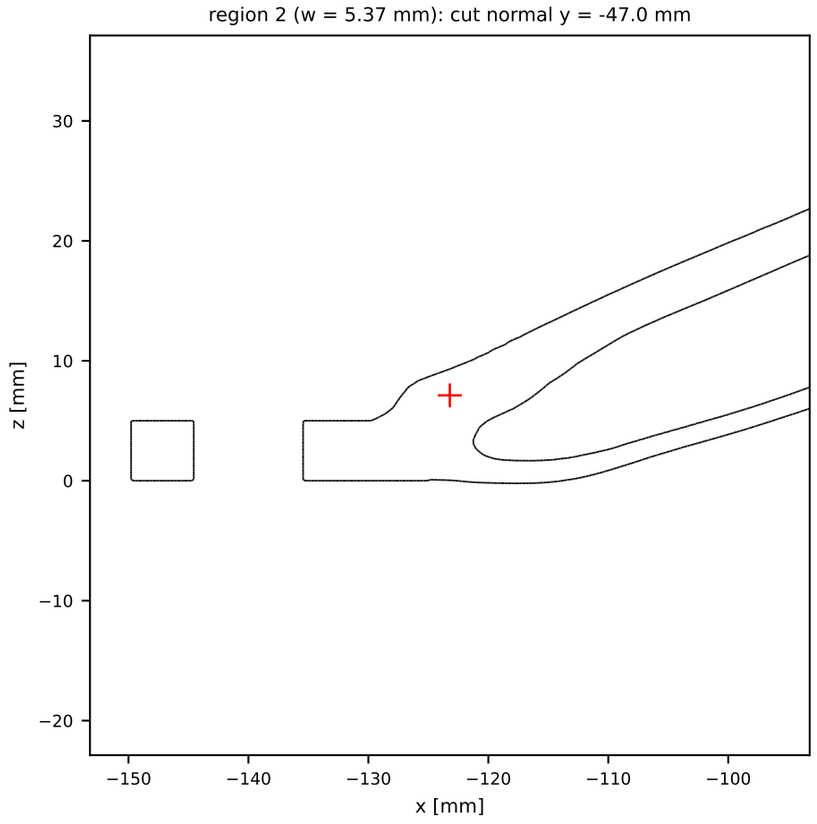
MBB: sections of the final design through the two thickest-wall regions (red +); +-30 mm window

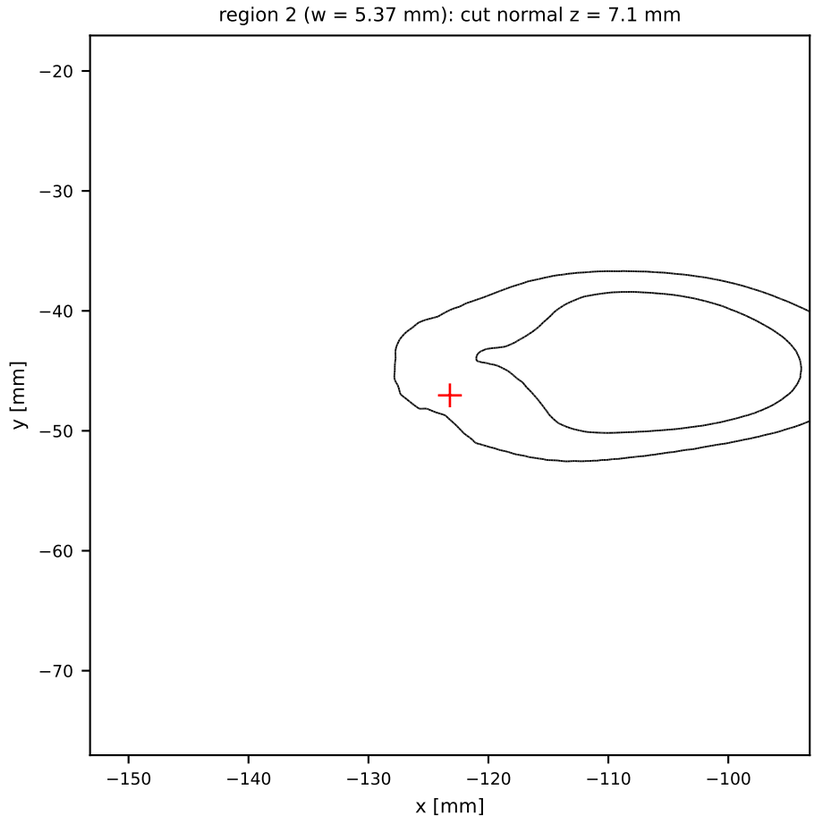






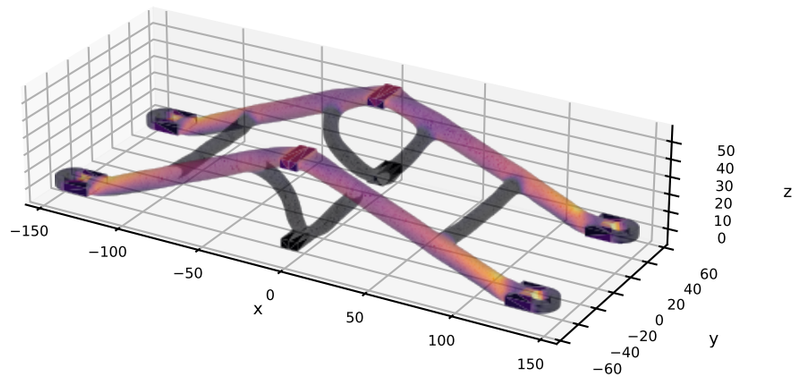




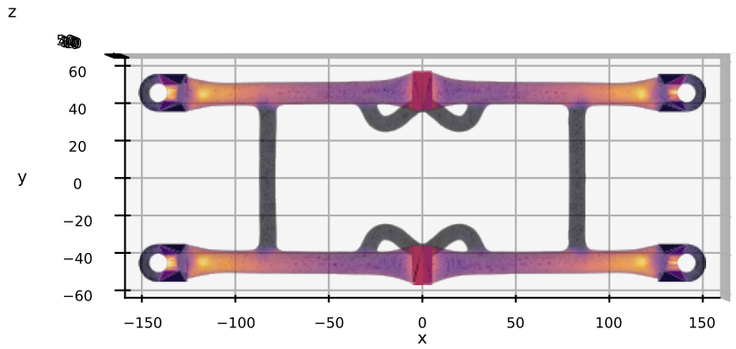


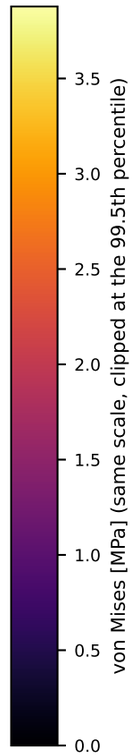
MBB: von Mises stress, original solid vs final design (peak solid 12.6, final 9.5 MPa; values from fields.vtp, nearest point)

solid: isometric

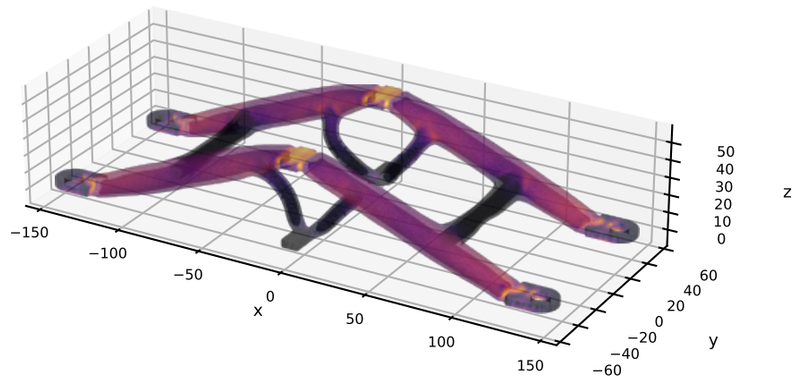


solid: top (+z)

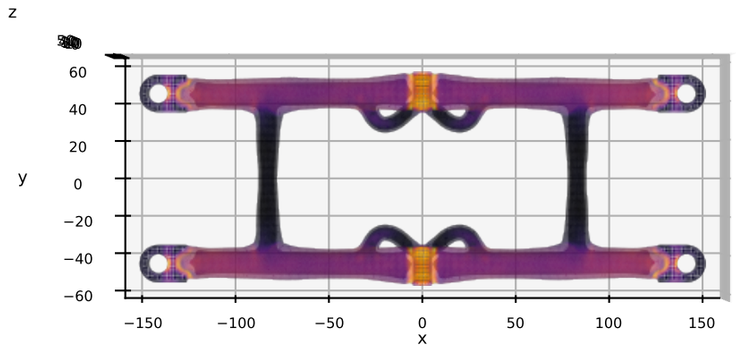


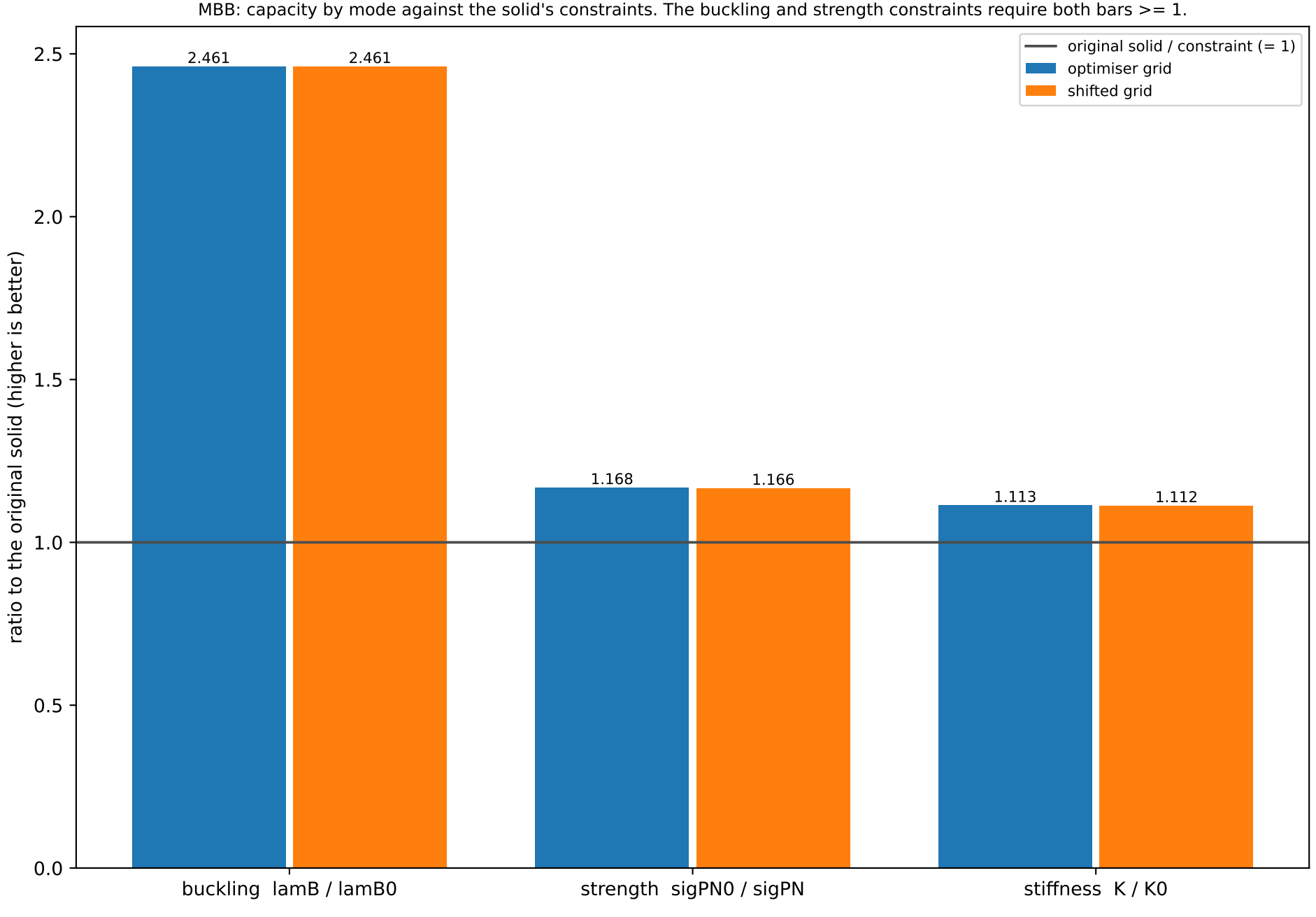


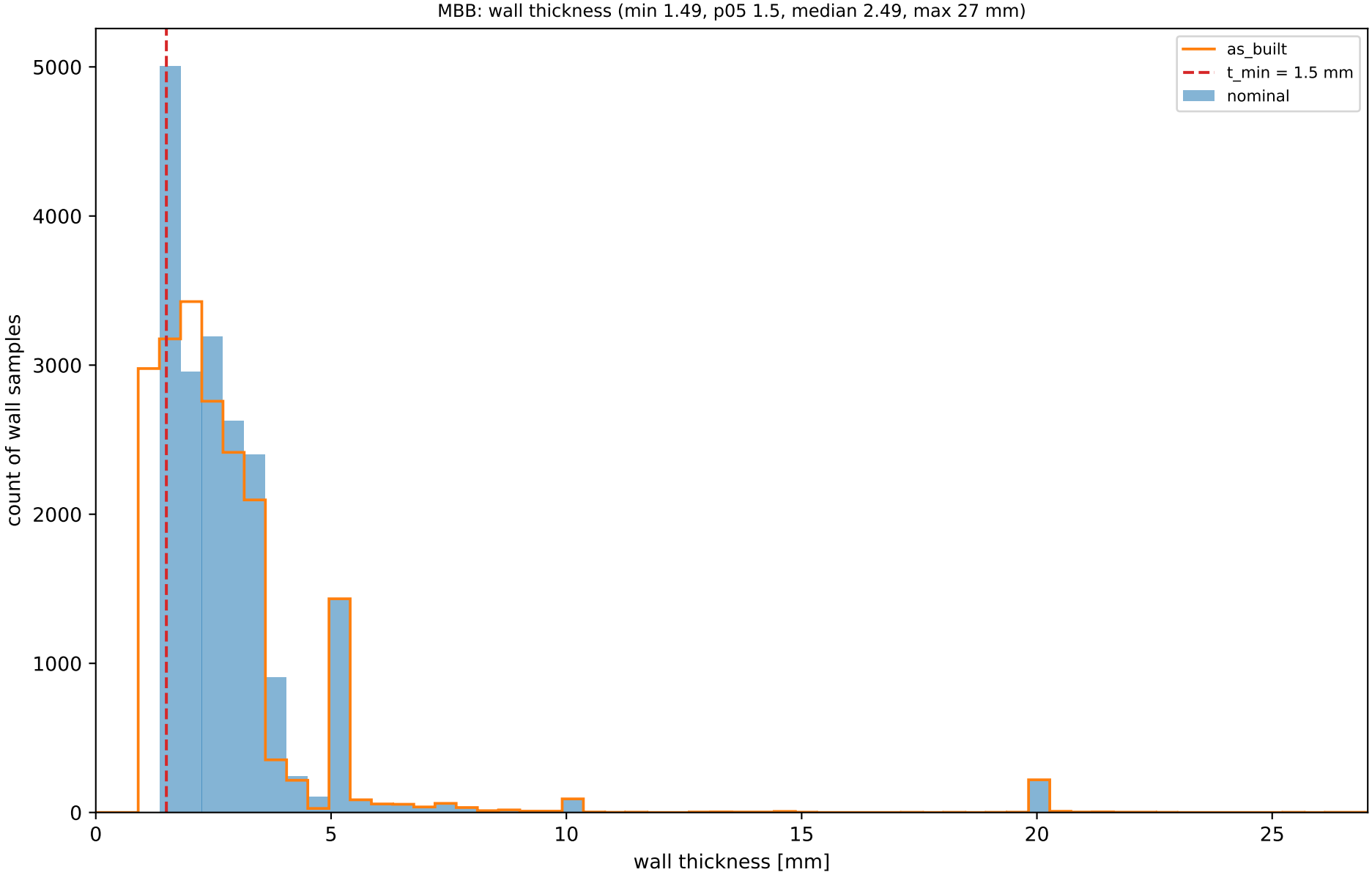
final design: isometric



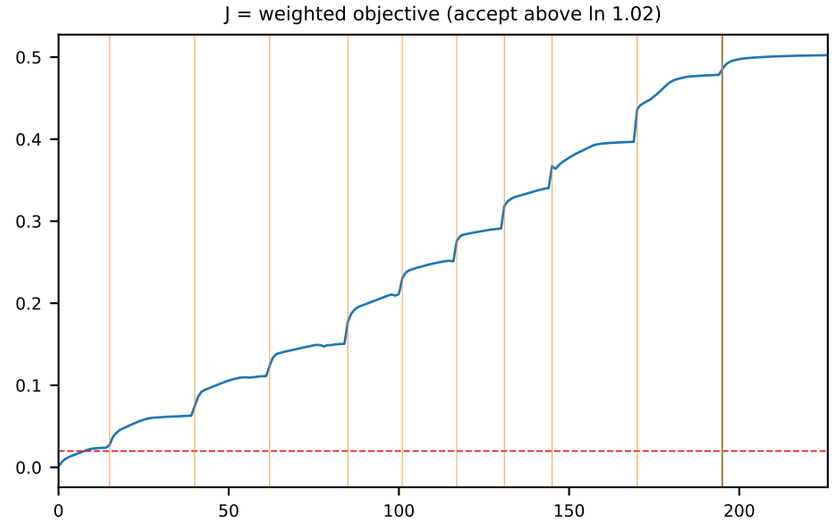
final design: top (+z)

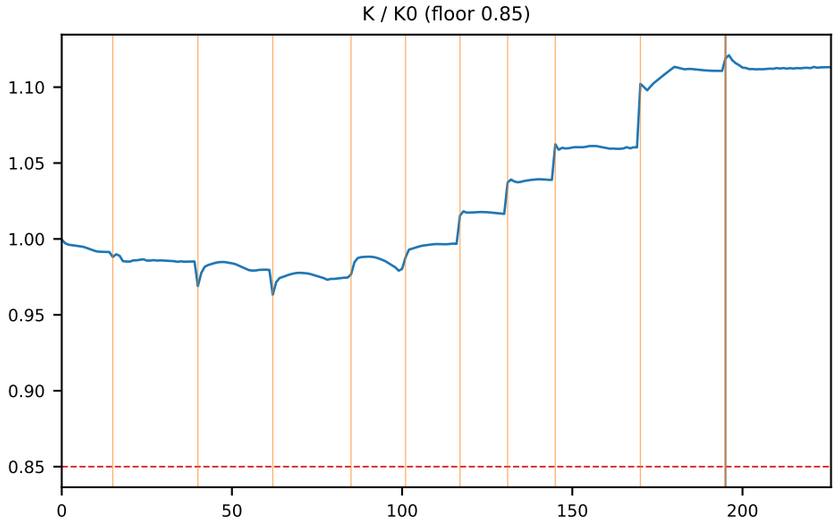


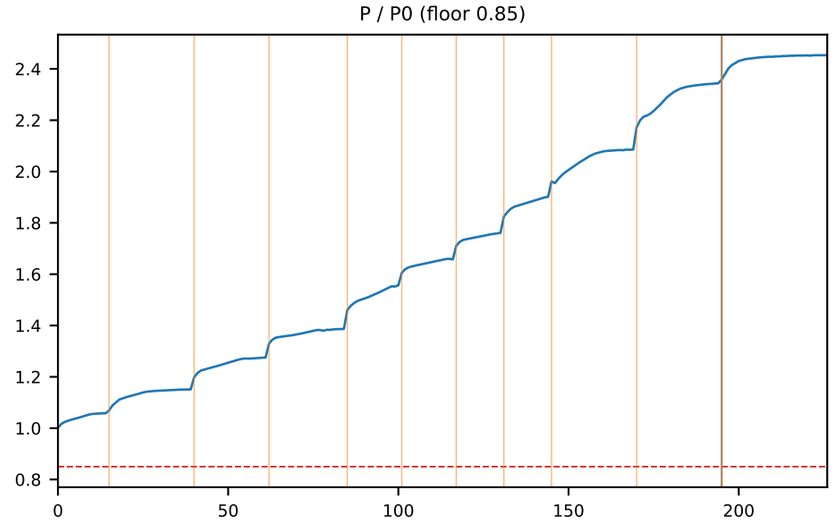

MBB: capacity by mode against the solid's constraints. The buckling and strength constraints require both bars >= 1.
original solid / constraint (= 1)
optimiser grid
shifted grid
ratio to the original solid (higher is better)
2.5
2.0
1.5
1.0
0.5
0.0
2.461
2.461
1.168
1.166
1.113
1.112
buckling lamB / lamB0
strength sigPN0 / sigPN
stiffness K / K0

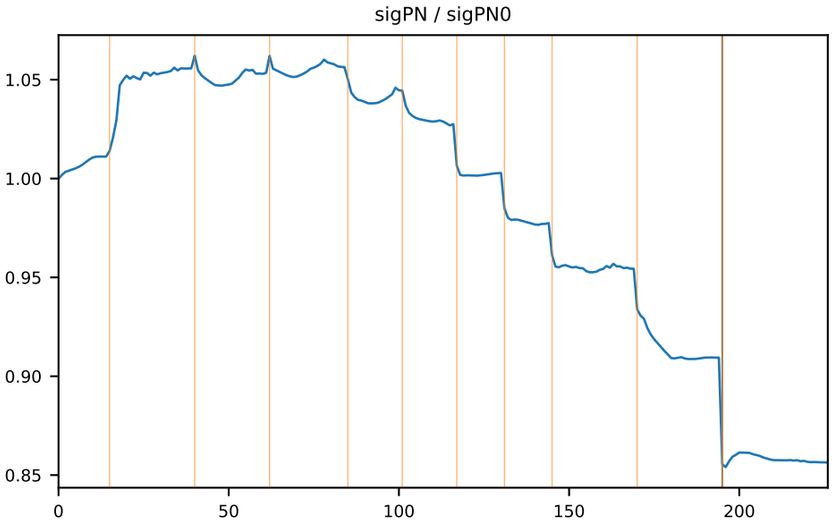

MBB: wall thickness (min 1.49, p05 1.5, median 2.49, max 27 mm)
as_built
t_min = 1.5 mm
nominal
count of wall samples
wall thickness [mm]
0
1000
2000
3000
4000
5000
0
5
10
15
20
25

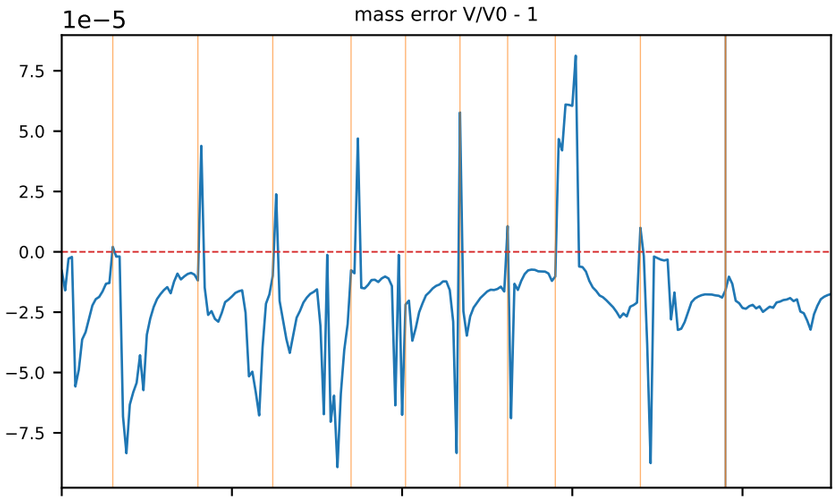
MBB: convergence history

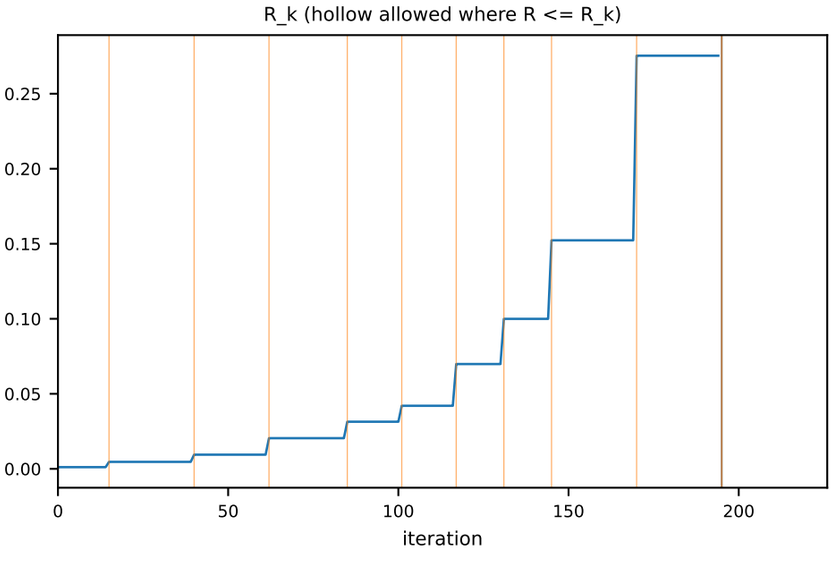


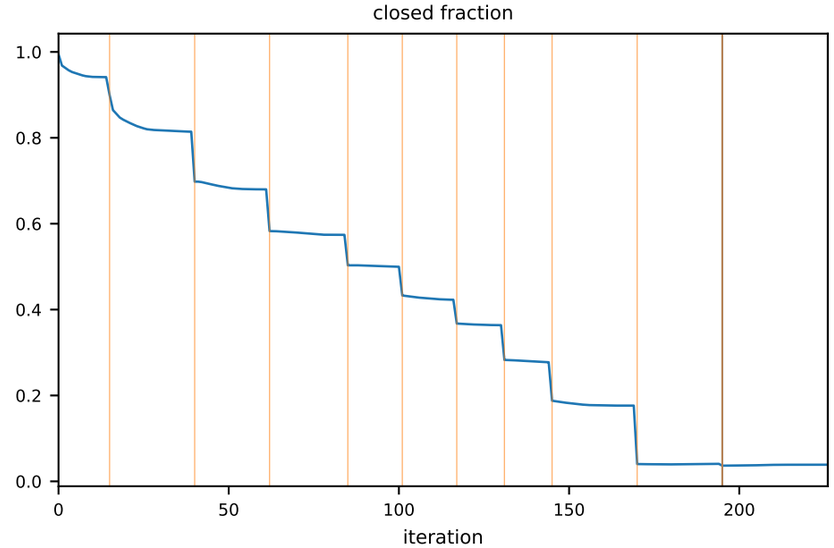


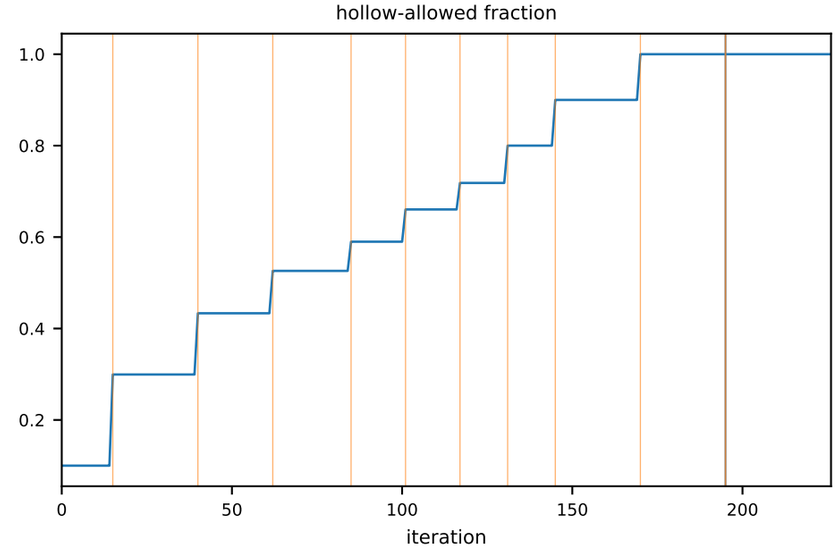








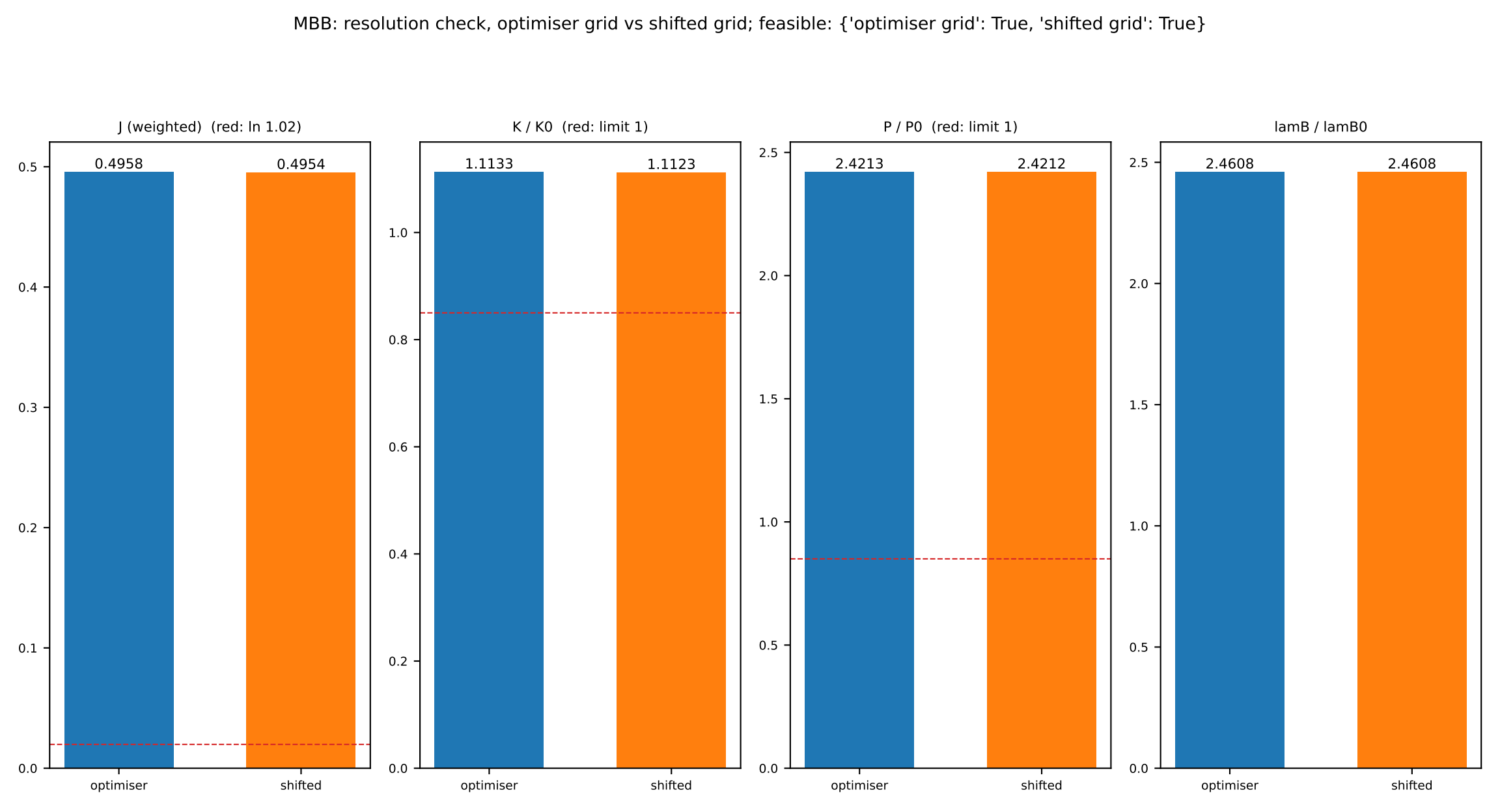


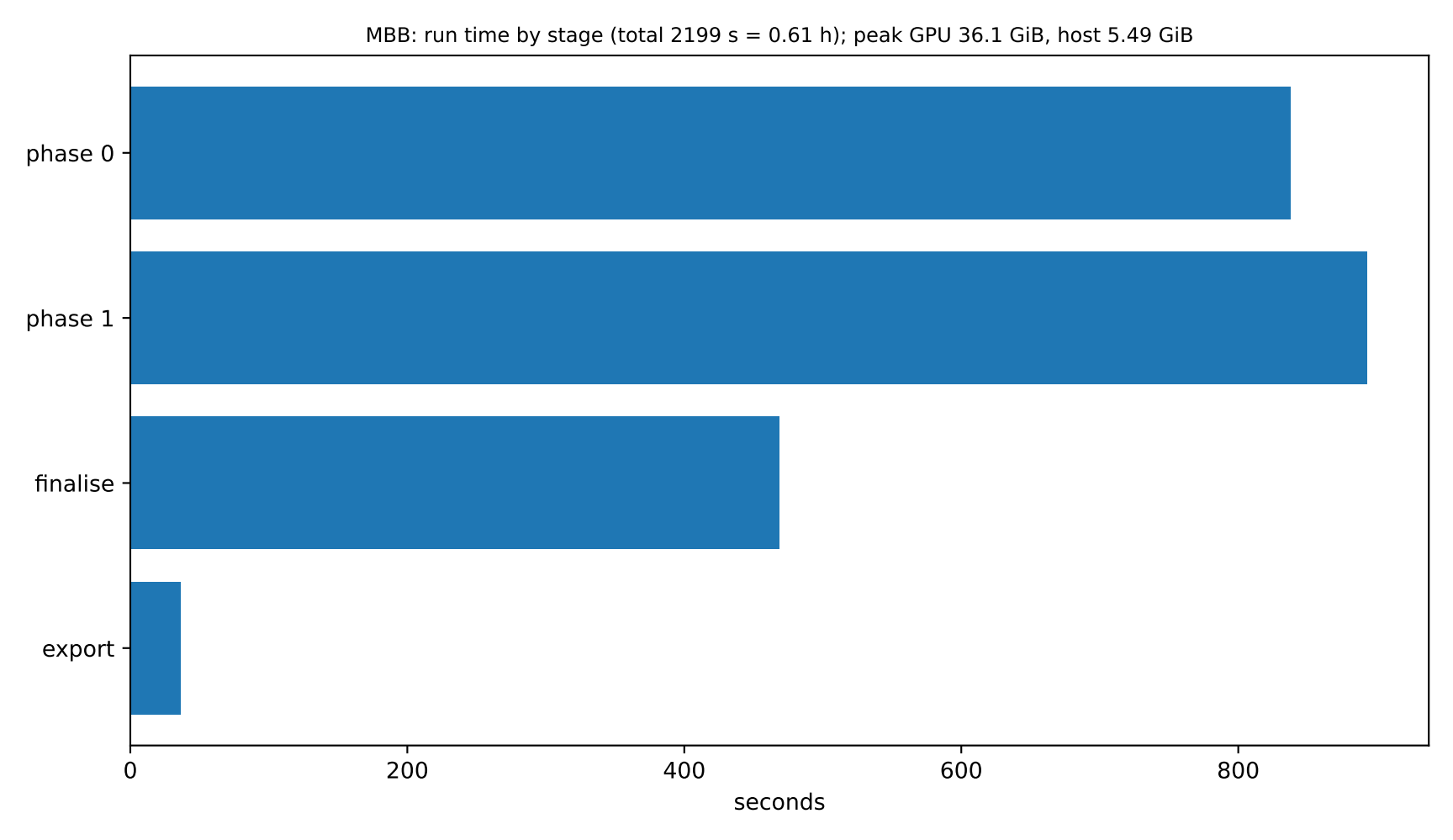


grey vertical: phase boundary
orange vertical: growth step (R_k rise)
iterations 227; growth steps 10
red dashed: constraint / acceptance level

```
enclosed voids: 2;  escape holes: 0;  voids without a hole: 2
  void 0: volume 29118 mm3, holes 0
  void 1: volume 29119 mm3, holes 0

Escape holes sit at the lowest-stress wall points of each void (outward direction in escape_holes.csv).
```


MBB: resolution check, optimiser grid vs shifted grid; feasible: {'optimiser grid': True, 'shifted grid': True}
J (weighted) (red: ln 1.02)
0.4958
0.4954
K / K0 (red: limit 1)
1.1133
1.1123
P / P0 (red: limit 1)
2.4213
2.4212
lamB / lamB0
2.4608
2.4608
optimiser
shifted


MBB: run time by stage (total 2199 s = 0.61 h); peak GPU 36.1 GiB, host 5.49 GiB
phase 0
phase 1
finalise
export
0
200
400
600
800
seconds